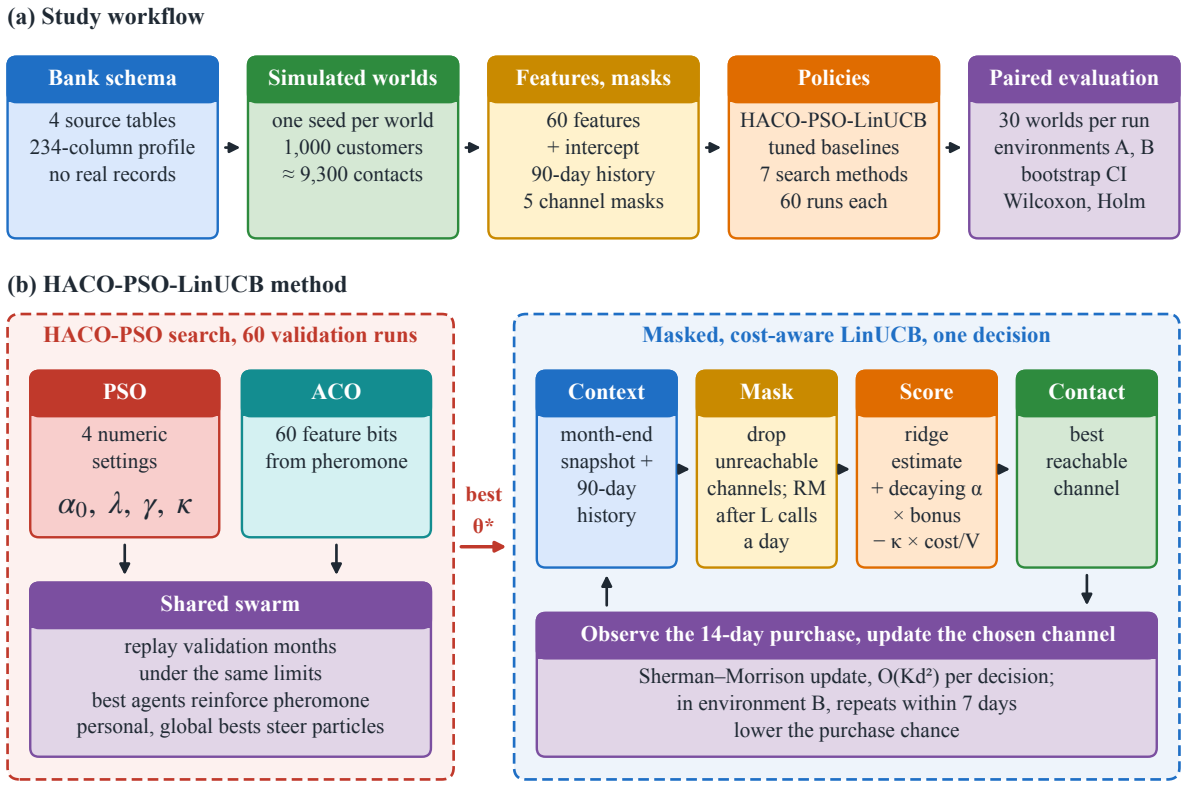

(a) Study workflow
Bank schema
4 source tables
234-column profile
no real records
Simulated worlds
one seed per world
1,000 customers
≈ 9,300 contacts
Features, masks
60 features
+ intercept
90-day history
5 channel masks
Policies
HACO-PSO-LinUCB
tuned baselines
7 search methods
60 runs each
Paired evaluation
30 worlds per run
environments A, B
bootstrap CI
Wilcoxon, Holm
(b) HACO-PSO-LinUCB method
HACO-PSO search, 60 validation runs
PSO
4 numeric
settings
$\alpha_0,\ \lambda,\ \gamma,\ \kappa$
ACO
60 feature bits
from pheromone
Shared swarm
replay validation months
under the same limits
best agents reinforce pheromone
personal, global bests steer particles
best
θ*
Masked, cost-aware LinUCB, one decision
Context
month-end
snapshot +
90-day
history
Mask
drop
unreachable
channels; RM
after L calls
a day
Score
ridge
estimate
+ decaying α
× bonus
− κ × cost/V
Contact
best
reachable
channel
Observe the 14-day purchase, update the chosen channel
Sherman–Morrison update, O(Kd²) per decision;
in environment B, repeats within 7 days
lower the purchase chance